

Fig. 2: Block diagram of SoC

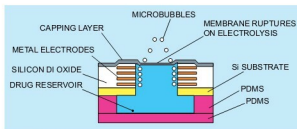


Fig. 3: Cross-section of reservoir

ted data stream is modulated by the OOK receiver and transmitted by a transmitter outside the body. The SoC implanted in the patient's body detects the wireless signal and demodulates it using an envelope detector. Demodulation by envelope detection significantly simplifies the system architecture and reduces power consumption as it does not require any frequency translation.

The demodulated signal is further sent to a microcontroller unit (MCU) for decoding. The decoded information contains the address of the reservoir to be opened up, so that the MCU can direct current into the electrodes of the appointed reservoir.

When the current is 'on,' the electrodes begin to generate microbubbles in the solution, which accumulate inside the sealed reservoir until the capping membrane is broken by the gas pressure. The power-on-reset regulator resets voltages of the MCU's registers at the system start-up before residual voltages result in incorrect system

behaviours. The regulator generates a stable supply voltage for the OOK receiver, while the clock generator is used to provide two different clocks for the MCU.

Cross-sectional view of a single unit of drug delivery system is shown in Fig. 3. The passivation layer formed by standard CMOS process is used as the membrane capping for the reservoir, which is opened up by the electrolysis approach.

As shown in Fig. 1, the drug-delivery chip is mounted on board. The PCB is further stacked on a Li-ion nanowire battery, which has high-density charge storage and is rechargeable. A planar inductor on the bottom of the package is used to receive the command signal and the wireless power from command OOK transmitter outside the body. The received power can be stored in the Li-ion battery, for a longer device lifetime. The drug-delivery device is implanted subcutaneously due to signal attenuation in the human body and limited size of the coupling inductor.

Drug reservoir fabrication

The drug-delivery SoC is fabricated by standard CMOS process followed by post-IC processing. After receiving the standard silicon chip with metal

electrodes from the foundry, post-IC micromachining is applied to each of the reservoir structures.

First, the standard chip is thinned to 100 μm by mechanical lapping of the substrate. Then photolithography is performed on the back side of the chip to define the positions and sizes of reservoirs. After hard baking the photoresist as the etching blocking layer, the unblock area is etched by inductively-coupled ion-enhanced plasma etching (ICP) to remove unwanted silicon, enabling formation of drug reservoir. Thereafter, silox vapour wet etching is done to remove the silicon dioxide filled between the electrodes. The electrodes thus exposed are now ready for electrolysis.

After back-side processing, post-IC processing is continued on front side. Passivation etching is done to create a thin capping layer of 200nm thickness in order to decrease the stiffness of the membrane. After this step, reservoirs complete with membrane and electrodes are formed, ready for drug filling.

Cross-section of the integrated reservoir is shown in Fig. 3. In order to increase the reservoir volume capacity, a polydimethyl-siloxane (PDMS) layer is bonded on the back side of the chip. Each reservoir is 210 μm long and 110 μm wide.

After PDMS bonding, the reservoirs are filled with the designated drug by an automatic dispenser. A total of eight reservoirs are made with total die size of $1.77 \times 1.4 \text{ mm}^2$. By bonding PDMS layer, the capacity of drug reservoir is increased from 5 to 200 nL. Another PDMS film is used to seal the bottom of the reservoir after drug filling. The total thickness of the drug delivery SoC is 0.7 mm. The dose of the drug may be adjusted by controlling the number of opened reservoirs. The reservoirs are designed to be opened up one at a time according to the received command.

On-chip electrolysis

The releasing mechanism of the system is based on electrolysis. The metal elec-